Till: a shop ledger you can trust

Till is the books and the stock for one merchant with more than one shop. Adaeze Studio has Ikeja and Surulere. A cashier sells. A manager counts. An admin opens a branch. Money is whole kobo. NGN 1.00 is 100 kobo. Nobody types a price on the sale. The shelf price is copied from the product.

1. A cash sale

One database transaction locks the product, then the stock. It takes the units off the shelf and writes the journal. If either part fails, both come back. The journal debits the Ikeja or Surulere cash account for what the customer paid, credits sales for the net, and credits VAT when the rate is not zero. VAT is inside the price. At 7.5 percent, 450000 kobo splits into VAT 31395 and net 418605. Net plus VAT equals the amount paid.

2. A discount

The cashier may send discount_minor, an integer in kobo, for the whole order. Zero is allowed. The discount must be smaller than the shelf total. VAT is worked out on what the customer pays, not on the shelf price. A 50000 kobo discount on a 450000 kobo item leaves 400000. VAT is 27906 and the net is 372094. The discount does not reduce the cost of the goods.

3. Store credit

A customer has a name, a phone, and a balance. A top-up takes cash into the drawer and raises that balance. A wallet sale spends the balance and takes the stock in the same transaction. The balance cannot go below zero. Account 2200, Customer wallet, is a liability: the shop owes the customer goods. The sum of the customer balances equals that account. A refund puts the money back on the balance and reverses the journal together.

This is not a wallet you can send to a bank or spend in another shop. Money that can leave the shop is stored value, and in Nigeria that needs a Central Bank licence. Till does not do that.

4. Average cost

Each product has avg_cost_minor. A receipt can name unit_cost_minor. The new average is the old stock times the old average, plus the new units times their cost, divided by the new total units, rounded down. The receipt debits Inventory (1200) and credits Goods received (2150). It does not empty the cash drawer. Paying the supplier later is a transfer from cash to goods received. A sale debits Cost of goods (5000) and credits Inventory for quantity times the average at that moment. Changing the average by hand, while units are still on the shelf, posts the difference to Inventory adjustment (5200). Raising the average credits that expense account, so 5200 is allowed to go negative.

5. A transfer

A transfer holds the units. It does not post a sale. The Paystack-style webhook is the only thing that posts it. The signature is the hex HMAC-SHA512 of the raw body. A bad signature changes nothing. The amount must match what the customer pays, after any discount. Cost is copied when the webhook arrives, because a receipt during the hold can change the average. The hold lasts 15 minutes unless you set TILL_HOLD_MINUTES. After that the units go back and no journal is written.

6. Why it stays one service

Stock and money commit together. If those lived in two services, one could succeed and the other could fail, and the books would lie. Kubernetes is a way to run many services. This laptop cannot run Docker, so it cannot run Kubernetes. One API process and one Postgres is the whole deployment. More capacity later means more copies of this same process, still on one database.

7. What you run

1. Postgres 16 on 127.0.0.1 port 5432, database till, user till, password till.

2. Copy .env.example to .env and run: go run ./cmd/tilld

3. Log in as admin@till.local with password change-me-now-12. That password is for the demo only.

4. Ikeja soap starts at 3 on hand with reorder 5, so it is low. Surulere soap is 20 and is not.

The sample average costs are written only the first time the database is empty. If the shop was seeded before cost existed, drop the database named till and start again. Do not drop till_test while tests are running.